# LIMIT PRESERVING ON OBJECTS IS STABLE UNDER COMPOSITION

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*Editorial note.* The closing 'We omit the verification that (limit-preserving) is commutative' is a routine diagram check; the sentence is retained verbatim and flagged for the reviewer.

*Source reference.* Original equation 06CU: sources/stacks-criteria/pinned-original-source.tex, source lines 570–570. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

**Lemma 0.1.** *Let $p : \mathcal{X} \to \mathcal{Y}$ and $q : \mathcal{Y} \to \mathcal{Z}$ be 1-morphisms of categories fibred in groupoids over $(Sch/S)_{fppf}$. If $p$ and $q$ are limit preserving on objects, then so is the composition $q \circ p$.*

*Proof.* This is formal. Let $U = \lim_{i \in I} U_i$ be the directed limit of affine schemes $U_i$ over $S$, let $z_i$ be an object of $\mathcal{Z}$ over $U_i$ for some $i$, let $x$ be an object of $\mathcal{X}$ over $U$, and let $\gamma : q(p(x)) \to z_i|_U$ be an isomorphism. As $q$ is limit preserving on objects there exist an $i' \geq i$, an object $y_{i'}$ of $\mathcal{Y}$ over $U_{i'}$, an isomorphism $\beta : y_{i'}|_U \to p(x)$, and an isomorphism $\gamma_{i'} : q(y_{i'}) \to z_i|_{U_{i'}}$ such that (06CU) is commutative. As $p$ is limit preserving on objects there exist an $i'' \geq i'$, an object $x_{i''}$ of $\mathcal{X}$ over $U_{i''}$, an isomorphism $\beta' : x_{i''}|_U \to x$, and an isomorphism $\gamma'_{i''} : p(x_{i''}) \to y_{i'}|_{U_{i''}}$ such that (06CU) is commutative. The solution is to take $x_{i''}$ over $U_{i''}$ with isomorphism

$$q(p(x_{i''})) \xrightarrow{q(\gamma'_{i''})} q(y_{i'})|_{U_{i''}} \xrightarrow{\gamma_{i'}|_{U_{i''}}} z_i|_{U_{i''}}$$

and isomorphism $\beta' : x_{i''}|_U \to x$. We omit the verification that (06CU) is commutative. □

*Date*: stacks-project:criteria.tex:local-snapshot-20261012; GNU FDL 1.2 or later.